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Module 2
Graded Assignment • 30 min

Due Nov 11, 11:59 PM IST

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 70%. We keep your highest score.

Next item →

1. Boyer-Moore: How many alignments are *skipped* by the bad character rule for this alignment? 1 / 1 point

Note: the number of skips is one less than the number of positions P shifts by. That is, if the pattern shifts by 2 positions, that's 1 alignment skipped.

Also note: the question is asking only about the alignment shown. Do not consider any other alignments of P to T in your answer.

```
1   T: GGCTATAATGCGTA
2   P:  TAATAAA
```

Nice work

2. Boyer-Moore: How many alignments are skipped by the good suffix rule in this scenario? 1 / 1 point

```
1   T: GGCTATAATGCGTA
2   P:  TAATTAA
```

Nice work

3. Boyer-Moore, true or false: for given P and T, it's possible that some characters from T will never be examined, i.e., won't be involved in any character comparisons. 1 / 1 point

Nice work

4. Consider a version of Boyer-Moore that uses only the bad character rule (no good suffix rule), and say our pattern P is a random string of 50% As and 50% Ts. In which scenario would you expect Boyer-Moore to skip the most alignments? 1 / 1 point

Nice work

5. The naive exact matching algorithm preprocesses: 1 / 1 point

Nice work

6. The Boyer-Moore algorithm preprocesses: 1 / 1 point

Nice work

7. In which of the these scenarios is an offline matching algorithm not appropriate? 1 / 1 point

Nice work

8. Say we have a k-mer index containing all 5-mers from T. We query the index using the first 5-mer from P and the index returns a single index hit. What can we say about whether P occurs in T? Assume T is longer than P and that P is at least 6 bases long. 1 / 1 point

Nice work

9. Say we have a k-mer index containing all k-mers from T and we query it with 3 different k-mers from the pattern P. The first query returns 0 hits, the second returns 1 hit, and the third returns 3 hits. What can we say about whether P occurs in T? 1 / 1 point

Nice work

10. Which of the following is not an "edit" allowed in edit distance: 1 / 1 point

Nice work